

DIELECTRIC CONSTANT KIT (FOR LIQUID)

OBJECTIVE

DETERMINATION OF DIELECTRIC CONSTANT of non-aqueous Liquid Samples.

Dielectric Constant Kit is a versatile instrument for measuring the dielectric constant of non-aqueous liquids. The apparatus is based on state of the art technology using physical concepts.



Block diagram of the apparatus is as shown in fig. 1.

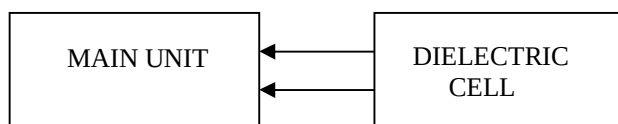


Fig 1

Main Unit has two terminals to which the di-electric cell has to be connected .

Main Unit Consists of

- i) Frequency Counter (4digit)
- ii) Oscillator : 500 KHz
- iii) Electronic Circuitry

Dielectric Cell Unit consists of

- i) Dielectric Cell (special grade SS) assembly
- ii) Beaker (100 ml)
- iii) Connecting Cable

WORKING PRINCIPLE:

A special circuit has been developed for oscillator producing stabilized 500 KHz wave. In this experiment dielectric cell is standardized using reference liquid having known dielectric constant by immersing the dielectric cell assembly in to reference Liquid. Then experimental liquid whose dielectric constant has to be determined is taken and assembly is immersed into liquid, resulting in change in oscillation frequency. From resulting shift, capacitance of cell in unknown liquid is calculated (C_X) . Dielectric Constant of unknown liquid is calculated using relation:

$$\epsilon = 1 + \frac{C_0 - C_X}{C_0 - C_{ST}} (\epsilon_{ST} - 1)$$

where C_0 = Capacitance of Air, C_{ST} = Capacitance of standard liquid, C_X = Capacitance of test liquid and ϵ_{ST} = dielectric constant of standard liquid.

Manufacturers:



MITTAL ENTERPRISES®

2151/T-7C, New Patel Nagar, New Delhi – 110008

Telefax: 011-25702784

Mobile: +91-9810681132, +91-9868532156

E-mail : mittalenterprises@bol.net.in, info@mittalenterprises.com

Website : <http://www.mittalenterprises.com>